Luke Serrano

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Experience

Land, Air, and Space Robotics (LASR) Lab Aug 2025 - Present

Computer Vision Lead College Station, TX

- Lead development and maintenance of computer vision capabilities on robotic testbeds in spacecraft emulation laboratory
- Created an object-detection–based stereo camera relative position tracking algorithm in Robot Operating System (ROS) to support downstream estimation and control
- Developing an optical relative pose estimation algorithm to facilitate precise simulated robotic satellite close-proximity docking and manipulation tasks
- Manage imagery datasets for object detection models, ROS Git repositories, and stereo camera integration on robotic platforms

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Sandia National Laboratories Year-Round Intern (Held Clearance) Jan 2024 - Aug 2025

Missile Guidance, Navigation, and Control (GNC) Engineer Albuquerque, NM

- Proved ability to work within a multi-language software framework, dive into mathematical theory, develop within a Git workflow, and contribute meaningfully to a GNC team
- Improved navigation regression test framework that stresses the robustness of current and future missile navigation algorithms by implementing the ability to playback flight sensor data
- Analyzed telemetry from IMU calibration procedure and C++ navigation software to create an independent Inertial Navigation System algorithm in an attempt to explain accumulation of velocity-error
- Assisted in the generation of optimal trajectories and creation of a thrust vector control autopilot to supply customer with impact results from an end-to-end Monte Carlo analysis

Missouri S&T Satellite Research Team (M-SAT) Aug 2022 - May 2025

Co-Chief Engineer and GNC Subsystem Member Rolla, MO

- Managed Git repositories and NASA's 42 simulator integration for close-proximity GNC team

Personal Projects

- Designed, implemented, and compared EKF/MEKF against Extended Gaussian Mixture Filter for lunar lander pose estimation utilizing inertial and landmark measurements (Python)(Fall 2025)
- Leveraged NASA's Godard Imaging and Analysis Navigation Tool to perform camera calibration, stellar, and relative optical navigation with NASA DAWN imagery (Python)(Spring 2025)
- Implemented an LQR optimal error-quaternion attitude controller for a spacecraft using Colorado Boulder's Basilisk astrodynamics simulation framework (Python)(Spring 2025)

Education

Texas A&M University Fall 2025 - Spring 2027
Pursuing M.S. in Aerospace Engineering **GPA: 4.0**
Optimal Estimation of Dynamic Systems, Advanced Dynamics , Orbital Mechanics, Optimal Control

Missouri University of Science & Technology Fall 2022 - Spring 2025
B.S. in Aerospace Engineering **GPA: 3.82**
Orbital Mechanics II; Orbit Determination and Estimation; Optimal Control

Skills

- Object-Oriented Programming
- C++ syntax, compilation, and linking
- Git Workflow
- Pub/Sub Messaging
- Mathematical Troubleshooting
- Python
- Linux Environment
- Simulink Simulation Development
- Self-Directed Learner